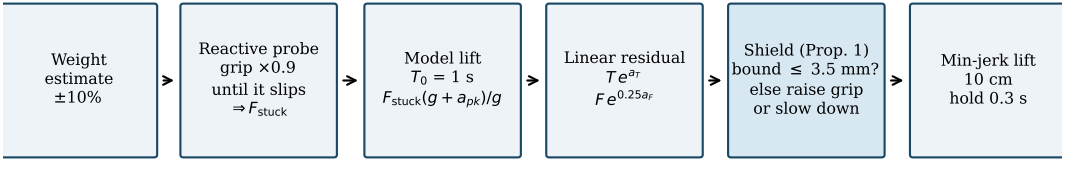
Weight estimate ±10%
Reactive probe grip ×0.9 until it slips ⇒ F_{stuck}
Model lift $T_0 = 1$ s $F_{\text{stuck}}(g + a_{pk})/g$
Linear residual $T e^{a_T}$ $F e^{0.25 a_F}$
Shield (Prop. 1) bound ≤ 3.5 mm? else raise grip or slow down
Min-jerk lift 10 cm hold 0.3 s